Exact Exchange in Levi–Perdew–Sahni Density Functional Theory

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In this work, we identified another exact term within the Levy–Perdew–Sahni formalism for the square root of the electron density. This term can be interpreted as the exact orbital-free exchange energy. Our analytical derivation relies on a specific choice of wave function describing a reference system of non-interacting boson-like fermions. This perspective preserves the conventional structure of Levy–Perdew–Sahni theory while rigorously defining the exact exchange term. This advancement enables the construction of orbital-free functionals containing exact exchange. Furthermore, our numerical study demonstrates that including the exact exchange in the Levy–Perdew–Sahni scheme can improve the accuracy of self-consistent calculations.

I. INTRODUCTION

The Levy–Perdew–Sahni theory (LPS) offers a rigorous formulation of density functional theory (DFT), determining the ground-state density of a many-electron system via a Schrödinger-like equation for the square root of the density

$$\left\{-\frac{1}{2}\nabla^2 + v_{\text{eff}}([n];\mathbf{r})\right\} n^{1/2}(\mathbf{r}) = \mu n^{1/2}(\mathbf{r}), \tag{1}$$

where $v_{\text{eff}}([n];\mathbf{r})$ acts as the multiplicative effective potential. The eigenvalue μ represents the chemical potential, which corresponds to the negative of the ionization energy in exact DFT [1]. This formulation of DFT contrasts with the Kohn–Sham (KS) DFT that is widely adopted for its effective compromise between accuracy and computational cost [2–4]. However, the LPS framework offers a distinct advantage since its computational cost scales near linearly with the electron number, N, comparing to the cubic scaling of KS DFT which limits its applicability to relatively small systems [5].

In original work, the authors identified the energy functional $E[n]$ containing exact terms for the kinetic, electron-nuclear, and classical Coulomb energies, leaving a part of the energy, $G[n]$, that must be approximated [1]. In this paper, we analytically derive another exact term in the LPS formalism that can be interpeted as the exact exchange functional. This exact term depends strictly on the electron density and does not depend on orbitals in contrast to the KS exact exchange [4], making it suitable for the LPS framework.

This advancement is particularly useful for the developement of accurate orbital-free approximations to $G[n]$, specifically to the exchange-correlation part of it. Currently, the LPS framework is limited to local and semi-local exchange-correlation functionals [6, 7] since the more accurate hybrid functionals require the exact KS exchange that is explicitly orbital-dependent [4].

In the following sections, we briefly review the LPS formalism, present the derivation of the LPS exact exchange functional, and provide numerical illustrations of its performance in self-consistent calculations comparing to the local LDA [8] and semi-local PBE [9, 10] exchange functionals from KS DFT.

II. REVIEW OF THE LPS THEORY

In LPS theory, the energy of interacting electrons $E[n]$ includes bosonic non-interacting kinetic energy (the von Weizsäcker functional [11])

$$T_{\text{W}}[n] = \int n^{1/2}(\mathbf{r}) \left(-\frac{1}{2}\nabla^2\right) n^{1/2}(\mathbf{r})\, d\mathbf{r}, \tag{2}$$

the electron-nuclear attraction energy

$$V_{\text{ext}}[n] = \int v_{\text{ext}}(\mathbf{r}) n(\mathbf{r}) d\mathbf{r}, \tag{3}$$

and the classical Coulomb energy

$$J[n] = \frac{1}{2}\int\int \frac{n(\mathbf{r})n(\mathbf{r}')}{|\mathbf{r}-\mathbf{r}'|} d\mathbf{r}d\mathbf{r}'. \tag{4}$$

The rest is approximated via universal functional $G[n]$ leading to the total LPS energy functional expression

$$E[n] = T_{\text{W}}[n] + V_{\text{ext}}[n] + J[n] + G[n]. \tag{5}$$

Given an approximation to $G[n]$, Eq. (1) can be solved with the effective potential defined as

$$v_{\text{eff}}([n];\mathbf{r}) = v_{\text{ext}}(\mathbf{r}) + \int \frac{n(\mathbf{r}')}{|\mathbf{r}-\mathbf{r}'|} d\mathbf{r}' + \frac{\delta G[n]}{\delta n(\mathbf{r})}. \tag{6}$$

Although the functional $G[n]$ can be approximated as a whole, this is customary to decompose it into kinetic and exchange-correlation components with the last borrowed from KS DFT [7]

$$G[n] = T_{\text{P}}[n] + E_{\text{xc}}^{\text{KS}}[n]. \tag{7}$$

The Pauli term, $T_{\text{P}}[n]$, is defined as a difference between the exact non-interacting electronic kinetic energy functional, $T_{\text{s}}[n]$ [4], and the von Weizsäcker term

$$T_{\text{P}}[n] = T_{\text{s}}[n] - T_{\text{W}}[n]. \tag{8}$$

One of the popular choices for $G[n]$ is the Thomas–Fermi–Dirac-λ-Weizsäcker (TFDλW) approximation [12–15]

$$T_{\text{P}}[n] = T_{\text{TF}}[n] + (\lambda - 1)T_{\text{W}}[n] \tag{9}$$

$$E_{\text{xc}}^{\text{KS}}[n] = E_{\text{x}}^{\text{LDA}}[n]. \tag{10}$$

In this model, $T_{\mathrm{P}}[n]$ is constructed as a linear combination of the TF kinetic energy functional [16, 17]

$$T_{\mathrm{TF}}[n] = \frac{3}{10}\left(3\pi^2\right)^{2/3}\int n^{5/3}(\mathbf{r})d\mathbf{r} \tag{11}$$

and the von Weizsäcker term, with a mixing parameter λ. $E_{\mathrm{xc}}[n]$ is approximated using the Dirac (LDA) exchange functional [8]

$$E_{\mathrm{x}}^{\mathrm{LDA}}[n] = \frac{3}{4}\left(\frac{3}{\pi}\right)^{1/3}\int n^{4/3}(\mathbf{r})d\mathbf{r}. \tag{12}$$

Alternatively, the Dirac exchange may be substituted by GGA or meta-GGA functionals, such as PBE [9, 10]. As well as the TFDλW kinetic functional can be replaced with more accurate semi-local or non-local approximations [5, 12].

A distinctive feature of the LPS formalism is the ability to utilize KS-like codes for kinetic functionals that lack an explicit Laplacian operator. Examples include the local TF functional (11) and the semi-local Tran–Wesołowski (TW) functional [18]. The implementation is achieved by subtracting the von Weizsäcker component from the Pauli term in Eq. (8) when constructing the effective potential (6) [19]. For the TF functional, this is equivalent to using the TFλW approximation with $\lambda = 0$. An analogous procedure applies to the TW functional.

III. DERIVATION OF EXACT EXCHANGE IN LPS DFT

The definition of the exact exchange functional in one-determinantal approximations, such as Hartree–Fock or KS DFT, relies on the explicit choice of the wave function and is given by

$$E_{\mathrm{x}}^{\mathrm{exact}}[n] = \langle\Psi_{\mathrm{min}}|\hat{V}_{\mathrm{ee}}|\Psi_{\mathrm{min}}\rangle - J[n]. \tag{13}$$

Here, $\hat{V}_{\mathrm{ee}} = \sum_{i<j}^{N}|\mathbf{r}_i - \mathbf{r}_j|^{-1}$ is the two-electron operator, and Ψ_{min} is the wave function that minimizes the total energy. Note that the two-electron operator is spin-independent and can be applied to particles of any arbitrary spin-statistics.

While the original LPS formalism was derived for bosons, the authors noted its potential generalization to other types of particles [1]. Therefore, to derive the exact LPS exchange, we consider a fictitious system of non-interacting boson-like fermions with spin degeneracy $M = 2s + 1$ equal to the particle number N. This perspective preserves the conventional structure of the LPS theory while yielding another exact term serving as the orbital-free exchange functional.

The one-determinantal non-interacting wave function for fermions with $M = N$ consists of N orthonormal spin-orbitals $\{\psi_i(\mathbf{x})\}$. Each spin-orbital is defined as a product of a spatial function $\sqrt{n(\mathbf{r})/N}$ and a spin function $\{\sigma_i(\omega)\}$. Since there are N distinct spin states, the Pauli exclusion principle allows single spatial orbital for all particles. Consequently, the antisymmetry is confined entirely to the spin part of the wave function

$$\Psi(\mathbf{x}_1, \ldots, \mathbf{x}_N) = \frac{\Phi_{\mathrm{HP}}(\mathbf{r}_1, \ldots, \mathbf{r}_N)}{\sqrt{N!}}\det\{\sigma_i(\omega_j)\}, \tag{14}$$

where the spatial part Φ_{HP} is the Hartree product [20].

Substituting Eq. (14) into Eq. (13) yields the exact exchange functional for LPS DFT

$$E_{\mathrm{x}}^{\mathrm{LPS}}[n] \equiv E_{\mathrm{x}}^{\mathrm{FA}}[n] = -\frac{1}{2N}\int\int\frac{n(\mathbf{r})n(\mathbf{r}')}{|\mathbf{r}-\mathbf{r}'|}d\mathbf{r}d\mathbf{r}'. \tag{15}$$

One can recognize it to be the Fermi–Amaldi functional [21]. The expression (15) is the sought-for exact orbital-free exchange functional. This result aligns with our previous findings [22, 23] that the FA functional is the exact exchange energy for particles of any spin when they share a single spatial orbital. The corresponding FA potential is given by

$$v_{\mathrm{x}}^{\mathrm{LPS}}([n];\mathbf{r}) = -\frac{1}{N}\int\frac{n(\mathbf{r}')}{|\mathbf{r}-\mathbf{r}'|}d\mathbf{r}, \tag{16}$$

where N is treated as a constant parameter [24–30].

The LPS energy functional of Eq. (5) can be fully recovered with the current choice of the wave function of Eq. (14). Kinetic, electron-nuclear, Coulomb self-repulsion, and the Fermi–Amaldi exchange terms are exact since they are the expectation values of the corresponding operators. The remaining term, $G[n]$, is an approximate functional needed to reconcile the boson-like fermion reference system with the true interacting electron density.

IV. NUMERICAL ILLUSTRATIONS

A. Implementation Details

To implement the iterative KS-like solution of Eq. (1) with the effective potential of Eq. (6), we adapted the open-source Psi4NumPy code [31, 32]. The density matrix was constructed exclusively from the lowest-energy eigenfunction normalized to N [1]. For open-shell systems, the α and β density matrices are generated from the lowest-energy eigenfunctions normalized to N_α and N_β, respectively. The spin-polarized forms of the approximate functionals constituting $G[n]$ were used.

A Core Hamiltonian initial guess was employed, and molecular integrals were computed analytically using a Gaussian-type basis set [20]. To obtain the self-consistent solutions, we used a combination of the direct inversion in the iterative subspace (DIIS) procedure [33] with Hartree damping [34]. The maximum length of the DIIS vector was varied from 3 to 9 depending on the system. For the calculations involving the TW kinetic functional, the DIIS was switched off as it makes the convergence

TABLE I. Atomic (H–Ar, Ca, Zn, and Kr) atomic SCF energy errors (relative to UHF/UGBS) for Fermi–Amaldi, LDA, and PBE exchange functionals combined with TFλW kinetic energy functionals using the basis set of 31 s-type basis functions for all atoms (see Sec. III B).

λ	1/9	1/6	1/5	1/3	1.0
			Mean Absolute Error (a.u.)		
LDA	24.93	7.81	1.07	28.69	106.73
PBE	26.43	9.22	0.79	27.46	105.82
FA	15.20	1.89	10.26	37.65	114.26
			Relative Mean Absolute Error (%)		
LDA	8.23	2.88	0.97	10.03	34.54
PBE	9.85	3.78	0.99	8.78	33.73
FA	5.95	0.87	2.86	11.93	36.00

unattainable. The damping factor was varied in the range $0.00-0.99$ and usually turned off closer to the convergence. The convergence criteria for the total energy and RMSD of the DIIS vector (or of the density matrix when the DIIS is not used) was both set to 1.0E-5 a.u.

B. Atomic Energies

In this section, we present benchmark calculations for a set of atoms of the first through third rows, as well as Ca, Zn, and Kr. We compare the exact LPS exchange (FA) (15) with the LDA exchange of Eq. (12), and the PBE exchange functional [9, 10]. To fully define $G[n]$, these exchange functionals are paired with the TFλW kinetic energy functional defined in Eq. (9) and the Tran–Wesołowski kinetic energy functional [18]. Specific values of λ were adopted from the literature: $\lambda = 1$, the original Weizsäcker value [11], known to substantially overestimate the kinetic energy [35, 36]; $\lambda = 1/3$, derived from a two-point trapezoidal discretization of the density matrix [15]; $\lambda = 1/5$, determined empirically for hydrogenic atoms [14, 37]; $\lambda = 1/9$, the coefficient for the second-order gradient expansion [4]; and $\lambda = 1/6$, which approximates the inclusion of the fourth-order correction [38]. For the TW functional, we utilized parameters fitted to the He and Ne ($\mu = 0.2335$, $\kappa = 0.8209$).

Since the atoms are minimized with spherical densities within LPS framework [12, 13], we use an ad hoc basis set consisting of 31 s-type Gaussians, $\exp(-\alpha r^2)$ ($\alpha \in [1.14 \times 10^7, 2.00 \times 10^{-2}]$), taken from the UGBS basis set [39–42]. Correspondingly, the large radial grid of 1000 points was used with the minimal number of spherical points (6) in the Lebedev grid.

Table I contains the mean absolute errors MAE $= \left(\sum_a \Delta E_a\right)/N_A$, where $\Delta E_a = |E_a^{\mathrm{LPS}} - E_a^{\mathrm{UHF}}|$, across $N_A = 10$ atoms, in a.u., and the relative mean absolute errors rMAE $= \left(-\sum_a \Delta E_a/E_a^{\mathrm{UHF}}\right)/N_A \times 100\%$ expressed as a percentage of the UHF/UGBS energy. In alignment with other authors [12, 13], Table I shows that PBE and LDA pure exchange functionals minimize errors with the empirical $\lambda = 1/5$, whereas the exact exchange (FA) strictly prefers the theoretical $\lambda = 1/6$. While PBE combined with $\lambda = 1/5$ yields the lowest mean absolute error (0.79 a.u.), the FA functional paired with $\lambda = 1/6$ achieves the superior relative accuracy (0.87%) across the test set. This confirms that the exact exchange functional is physically compatible with the fourth-order gradient expansion, unlike standard DFT approximations which rely on error cancellation with the $\lambda = 1/5$ model. This conclusion is further strengthened by the performance of the theoretical second-order coefficient ($\lambda = 1/9$), where FA significantly outperforms both LDA and PBE, confirming that the exact exchange functional is far more compatible with the gradient expansion series than standard DFT approximations.

Interestingly, in the SCF calculations, the highly-accurate Tran–Wesołowski kinetic functional yields higher errors for all exchange functionals compared to the optimal TFλW models. This observation suggests that the TW functional may not be well-suited for the self-consistent densities lacking the shell structure, as it was developed and parametrized for atomic densities exhibiting shell structure [18]. For post-SCF calculations, the TW PBE approximation applied to the UHF/UGBS densities (H–Ne) yields MAE = 0.07 a.u. and rMAE = 0.60%, respectively. The TF0.2W PBE model yields MAE = 2.93 a.u. and rMAE = 8.31% on the same set of atomic densities. Clearly, the TW functional is superior for post-SCF calculations, but its performance degrades in self-consistent calculations within the LPS framework.

Table II summarizes the performance of the FA, LDA, and PBE exchange functionals combined with TF$\frac{1}{6}$W kinetic functional for each exchange functional on a set of closed-shell atoms He through Kr. The basis set and numerical grid are the same as in Table I. Comparing to Table I, MAE values increased significantly for all tested exchange functionals which is the consequence of the larger magnitute of the total energies in Table II. Conversely, rMAEs are decreased for all models indicating the lower-

TABLE II. SCF atomic energies (a.u.) for closed-shell atoms from He to Kr calculated using TF$\frac{1}{6}$W model for LDA, PBE, and FA exchange functionals. Calculations are done with the same basis set and grid as in Table I. The MAE (a.u.) and rMAE (%) are with respect to RHF/UGBS values.

Atom	TF$\frac{1}{6}$W LDA	TF$\frac{1}{6}$W PBE	TF$\frac{1}{6}$W FA	RHF/UGBS
He	−2.9512	−3.0971	−3.0195	−2.8617
Be	−15.0404	−15.3889	−14.6605	−14.5730
Ne	−132.5089	−133.5737	−128.2917	−128.5471
Mg	−204.5407	−205.8645	−198.3054	−199.6146
Ar	−537.2088	−539.3465	−522.9289	−526.8175
Ca	−690.3963	−692.8150	−672.8085	−676.7582
Zn	−1812.1925	−1816.0679	−1773.7277	−1777.8481
Kr	−2795.9146	−2800.6966	−2741.6652	−2752.0549
MAE	13.96	15.97	3.02	–
rMAE	2.42	3.69	1.11	–

ing of the relative error with increasing atomic number. For both MAE and rMAE values, the FA functional outperforms both LDA and PBE significantly when paired with TF$\frac{1}{6}$W kinetic functional.

C. Chemical Potentials and Atomic Densities

Table III summarizes the chemical potentials μ of the LPS equation (1) for closed-shell atoms He through Kr calculated using TF$\frac{1}{6}$W model for each exchange functional.

TABLE III. Chemical potentials (a.u.) for closed-shell atoms from He to Kr calculated using TF$\frac{1}{6}$W model for LDA, PBE, and FA exchange functionals. The MAE (a.u.) and rMAE (%) are with respect to RHF/UGBS values.

Atom	TF$\frac{1}{6}$W LDA	TF$\frac{1}{6}$W PBE	TF$\frac{1}{6}$W FA	RHF/UGBS
He	−0.0671	−0.0812	−0.3135	−0.9180
Be	−0.0698	−0.0818	−0.2724	−0.3093
Ne	−0.0725	−0.0822	−0.2314	−0.8504
Mg	−0.0730	−0.0823	−0.2248	−0.2530
Ar	−0.0738	−0.0824	−0.2119	−0.5910
Ca	−0.0740	−0.0825	−0.2089	−0.1955
Zn	−0.0748	−0.0826	−0.1983	−0.2925
Kr	−0.0751	−0.0826	−0.1941	−0.5242
MAE	0.42	0.41	0.26	–
rMAE	80.31	77.80	40.98	–

The chemical potentials obtained with the FA functional are consistently closer to the RHF/UGBS reference values compared to those from LDA and PBE functionals. Overall, the FA functional demonstrates almost a two-fold improvement in accuracy for chemical potentials over both LDA and PBE functionals with the smallest errors of 0.0369, 0.0282, and 0.0133 Hartrees for the alkaline earth metals (Be, Mg, and Ca, respectively).

According to the LPS theory, the solution density assymptotically behaves like

$$n(\mathbf{r}) \sim \exp -2\sqrt{-2\mu} r. \tag{17}$$

This implies that the bigger magnitude of the chemical potential is associated with a faster decay of the density. To validate this, we examine a self-consistent radial distribution functions for He in Fig. 1. All calculations used the basis set with 31 s-type basis functions described above.

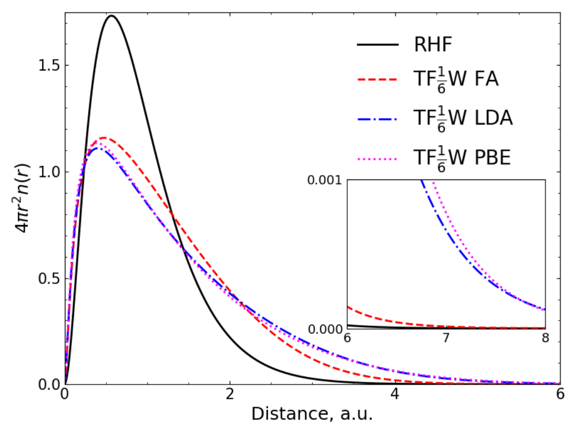

FIG. 1. He SCF radial distribution function $4\pi r^2 n(\mathbf{r})$ for FA, LDA, and PBE exchange functionals combined with TF$\frac{1}{6}$W kinetic energy functional. The RHF reference is included for comparison.

In alignment with Table III and Eq. (17), Fig. 1 shows that the FA density decays more rapidly than either the LDA or PBE densities, but slower than the RHF one. Comparison to the RHF density using Eq. (17) is valid for He, because it is a one-orbital system. For a system with more than one orbital, like Be (Fig. 2), Eq. (17) does not apply to the RHF density. We still put it on the plot to show that, as expected, the choice of exchange functional does not fundamentally alter the LPS density structure, i.e., the FA exchange does not recover atomic shell structure, which remains absent due to the monotonic nature of the TFλW kinetic model. However, Eq. (17) applies to the LPS densities in Fig. 2, showing the more rapid asymptotic decay in the FA density is consistent with Table III. The more rapid assymptotic decay has been reported before within statistical atomic theory [43].

Similar assymptotic behavior is observed when the Tran–Wesołowski kinetci functional is combined with the exchange functionals (Fig.3 and Fig.4).

D. Dissociation Profiles

Diatomic calculations employed the basis set containing 13 s- and 4 p-functions taken from the UGBS basis set

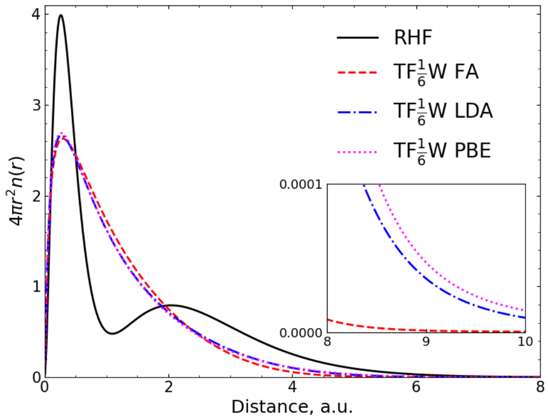

FIG. 2. Be SCF radial distribution function $4\pi r^2 n(\mathbf{r})$ for FA, LDA, and PBE exchange functionals combined with TF$\frac{1}{6}$W kinetic energy functional. The RHF reference is included for comparison.

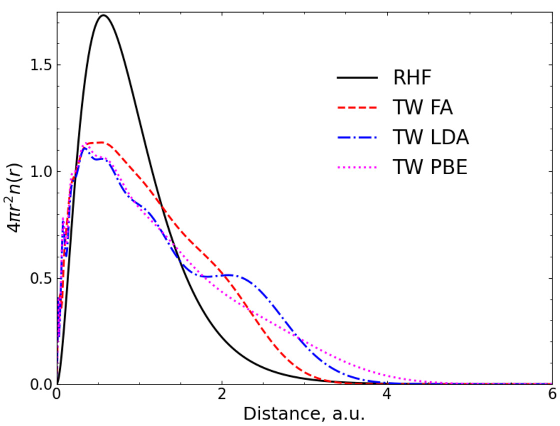

FIG. 3. He SCF radial distribution function $4\pi r^2 n(\mathbf{r})$ for FA, LDA, and PBE exchange functionals combined with TW kinetic energy functional.

($\alpha_s \in [9.34{\times}10^2, 7.67{\times}10^{-2}]$ and $\alpha_p \in [5.76{\times}10^{-1}, 3.92{\times}10^{-2}]$). Correspondingly, a radial grid of 1000 points and spherical grid of 974 points was used. In difficult convergence cases, the number of points was reduced to 350 and 74, respectively, to obtain an initial guess density for the larger grid calculations.

Figure II shows the H_2 dissociation curves computed using the same models as in the previous section for density plotting. An RHF curve with the same basis set is included as the reference. Although spin-polarized functionals were used, our LPS implementation lacks a symmetry-breaking mechanism. As such, it converges to the spin-symmetric solution, making RHF the appropriate spin-symmetric reference rather than the broken-symmetry UHF solution.

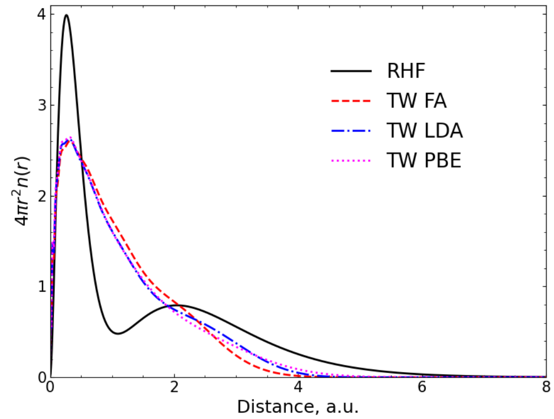

FIG. 4. Be SCF radial distribution function $4\pi r^2 n(\mathbf{r})$ for FA, LDA, and PBE exchange functionals combined with TW kinetic energy functional.

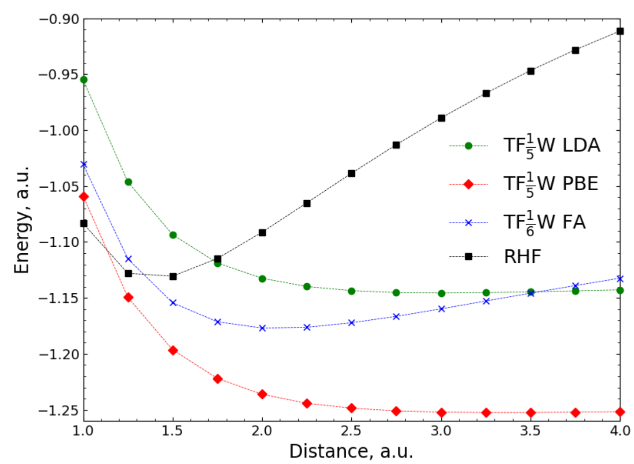

FIG. 5. H_2 SCF potential energy surface for Fermi–Amaldi, LDA, and PBE exchange functionals combined with TF$\frac{1}{6}$W and TF$\frac{1}{5}$W kinetic energy functionals, respectively. Calculations were performed with the basis set containing 13 s- and 4 p-functions (see Sec. III D). Self-consistent calculations were performed only at the marked points. The connecting curves are interpolations.

As shown in Figure II, the FA functional exhibits better long-range behavior and a distinct potential well unlike both the LDA and PBE functionals. The FA functional also yields an equilibrium distance, r_e, significantly closer to the RHF reference, e.g., 2.00 a.u. against 3.00 and 3.25 a.u. for LDA and PBE, respectively. The FA binding energy estimate, $E(4\,\text{a.u.}) - E(r_e) = 0.0134$ a.u., is closer to the RHF reference (0.2191 a.u.) compared to 0.0028 and 0.0007 a.u. for LDA and PBE, respectively.

V. CONCLUSION

In this work, we have reinterpreted the Levy–Perdew–Sahni (LPS) scheme to demonstrate that the Fermi–Amaldi functional is the orbital-free exact exchange functional within the LPS framework. By utilizing a reference system of non-interacting boson-like fermions, where all particles share a single spatial orbital, we showed that the conventional LPS energy functional [1] is fully recovered and another exact functional is easily defined.

The Fermi–Amaldi functional exhibits several highly desirable properties for an exchange functional. It is self-interaction free for one-orbital systems [22, 44], possesses the correct long-range asymptotic behavior [44], and exhibits the necessary derivative discontinuity at integer electron numbers N [24]. Furthermore, unlike the exact Kohn–Sham exchange potential, the Fermi–Amaldi potential is strictly multiplicative. Recently, we also extended the FA functional to fractional electron number [24]. In this work we also showed that the FA functional can be more accurate than LDA and PBE in self-consistent calculations. Collectively, these features identify the FA functional as a promising building block for future developments in LPS DFT.

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